



Santiago Aguilar Mello

Computer Science (ITC) · Embedded Systems & IoT

Mexico City, Mexico · Open to remote and relocation

Spanish & Portuguese (native) · English (C1) · Dutch (A1)

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Profile

Computer Science student at Tec de Monterrey working in embedded and IoT systems, in C++, Python, and ESP32. Shipped a live website for a paying freelance client and an MQTT-based smart parking prototype. IB diploma from the Netherlands.

Technical Skills

Programming: C++, Python, JavaScript, HTML/CSS
Embedded: ESP32 (Arduino / C++), MQTT, ThingSpeak API
Tools: Git/GitHub, Linux/Bash, VSCode, MySQL

Education

Tecnológico de Monterrey (ITESM) 2024 – Present
B.S. in Computer Science and Technology. Relevant coursework: Data Structures, OOP (C++), Embedded Systems & IoT, Databases, Software Engineering, Web Development.

Rotterdam International Secondary School (RISS), Netherlands 2022 – 2024
International Baccalaureate Diploma: 33/45. Awards: Excellence in English, high academic performance, leadership. Head of Charity: led campaigns including “Buy a Tulip, Help a Girl” and Red Cross disaster relief for Turkey & Syria.

Technical Projects

DT Construct ICS — Freelance Client Website dte-ingenieria.com
• Designed and shipped a live multi-page website for a Mexican construction company, hand-coded in HTML, CSS, and JavaScript with interactive Leaflet maps and a six-section services catalog.

IoT Smart Parking System — ESP32, MQTT, Python github.com/sant-mell/smart-parking-iot
• Built an ESP32 prototype with occupancy detection, automated barrier control, and MQTT telemetry streamed to a ThingSpeak dashboard.

Parallel DFA Syntax Highlighter — Python github.com/sant-mell/parallel-syntax-highlighter
• Wrote a DFA lexer by hand, explicit transition table and no regex, then parallelised it into a syntax highlighter: 6.2x the sequential baseline on 16 cores, over 60 files and 8 MB of Python.

The Fool’s Descent — JavaScript, Node/Express, MySQL github.com/sant-mell/videoGame-TC2005B.501
• Roguelike deckbuilder with a custom engine, a REST API, and a relational schema. Mid-duel state is checkpointed as JSON so a run resumes exactly where it left off.

Data Structures in C++
• Implemented BSTs, AVL trees, heaps, queues, hash logs, BFS/DFS, and Dijkstra, including full complexity analysis.

Work Experience

English Teacher — Pro English BV (Engels voor Bengels), Rotterdam Jul 2025 – Aug 2025
Taught English to 165 children (ages 4–12) using immersive, play-based methods.

Python Instructor (ages 12–16) — Logaritmia MX Feb 2025 – Jul 2025
Designed and taught a Python curriculum built around interactive coding challenges and mini-games used directly in class.

Volunteer & Peer Mentor — Tec de Monterrey 2024 – Present
Guided new students through their first year, academically and socially.

Interests: theatre and public speaking.